## NAME

delta − differences between two files

## SYNOPSIS

**delta** [*options*] *file1 file2*

## DESCRIPTION

**delta** computes the Longest Common Subsequence (LCS) of *file1* and *file2*, and prints their differences as an edit script. With **−l** it prints only the LCS length.

**delta** is a simplified companion of **compare**(1): it always reports differences (there is no correspondence mode), it always uses the Hunt−Szymanski algorithm, and it compares records rather than code points.

Each difference is introduced by a header line:

<<<<< n *file1* [*begin*:*end*] <<<<<
(records present only in *file1*)
>>>>> n *file2* [*begin*:*end*] >>>>>
(records present only in *file2*)

where *n* is the one-based difference number and *[begin:end]* the half-open range of records involved. A final line reports the LCS length, followed (when compiled with duration reporting enabled) by a timing line such as *Hunt-Szymanski: 28 operations in 9 µs @3.111 MHz*.

Each input line is one record; with **−w** each whitespace-separated word is one record.

## OPTIONS

**−b** Ignore whitespace: collapse each run of whitespace to a single space before comparing, and expand the reported spans back to the original positions.

**−E** Print the host environment (locale, processor, and memory) and exit.

**−h** Print a usage synopsis and exit.

**−i** Perform a case-insensitive comparison.

**−j** *join* Coalesce nearby differences that lie within *join* records of each other.

**−l** Report only the LCS length; skip reconstructing the differences.

**−p** *prefix*
Print *prefix* lines of context before each difference.

**−s** *suffix*
Print *suffix* lines of context after each difference.

**−t** *tabsize*
Set the tab size used to expand tab characters in the parallel **−y** p output (default 8).

**−v** Enable verbose output: print the LCS length, the start and finish wall-clock times, and a timing summary; by default only the differences are shown.

**−V** Print the version and exit.

**−w** Split each line into whitespace-separated words and compare word by word.

**−W** *width*
Set the per-side content width used by the parallel **−y** p format (default 72, minimum 5). Content is truncated to this width, so the total output line is 1 + 2 x width characters.

**−y** *s*|*p* Select the output format: **s** (series, the default: one side under the other) or **p** (parallel: side-by-side columns).

The **−h**, **−V**, and **−E** switches are early-exit switches: each prints its report and terminates without comparing files.

## EXIT STATUS

**0** Success (including **−h**, **−V**, and **−E**).

**1** Error: a command-line parse failure, a missing file, or an algorithm failure.

## ENVIRONMENT

The locale (in particular **LANG**, **LC_ALL**, and **LC_NUMERIC**) controls the thousands separator used in the reported counts and the locale name shown by **−E**. On a system whose locale defines digit grouping, counts are printed with comma separators (for example **LCS Length = 4,591,960**); otherwise they are printed ungrouped.

## EXAMPLES

Compare two files line by line and show the differences:

```
delta old.txt new.txt
```

Compare two files word by word, ignoring case:

```
delta −i −w a.txt b.txt
```

Report only the LCS length of two files:

```
delta −l a.txt b.txt
```

## SEE ALSO

**compare**(1), **diff**(1), **cmp**(1)